

Task-Driven Object Detection

With Semantic Distillation and Dynamic Gating using Accelerator on FPGA for Embedded Applications

An Assistive Vision Framework for Inclusive Education and Interactive Learning Analysis.

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01. Problem Statement: The Accessibility Gap

In inclusive learning environments (Labs/Art Studios), students with **visual impairments/visual search deficit** face a Semantic Barrier when using standard AI tools.

The Functional Challenge:

- **Label Dependency:** Traditional detectors require exact names ("Is there a glass?"). A student often needs an object based on its **Function** ("Is there something I can pour into?").
- **Cognitive Load:** Forcing a student to guess a list of synonyms (bottle, tumbler, cup) degrades the focus on the actual learning task.
- **Cluttering Confusion** A reduced ability to items within a cluttered or distracting environment. ADHD, Parkinson's, dyslexia, or spatial attentional issues
- **Educational Exclusion:** Lack of real-time, intent-based feedback prevents independent participation in STEM and Art experimentation.

01. Problem Statement : Hardware Barrier

Conventional detection (YOLO/SSD) is **Task-Agnostic** as it identifies all 80 COCO classes blindly, leading to computational waste on edge devices.

The Research Challenges:

1. **Contextual Irrelevance:** Standard models cannot prioritize objects based on goals (e.g., "Pouring").
2. **The VLM Weight Gap:** Large Vision-Language Models (CLIP) are too massive for the **VEGA RISC-V** processor.
3. **The Memory Bottleneck:** Standard **Attention** math ($O(N^2)$) exceeds the BRAM limits of the **Genesys-2 FPGA**.

Objective: To build a **Task-Driven** system that selectively filters features using **Dynamic Gating**, trained via **Knowledge Distillation** for **INT8 Execution**.

02. Theme Justification: SDG 4 Quality Education

Sub theme : Digital Identity and Learning Analysis.

Learning Analysis via Functional Intent:

- **Intent-Based Interaction:** Instead of analyzing "what" a student sees, we analyze the **Functional Context** of their learning.
- **Adaptive Support:** By observing the Task Vectors a student uses most frequently, educators can analyze a student's spatial reasoning and experimental flow in complex lab environments.

02. Literature Survey (fake atp)

Focus: Moving from General Detection to Task-Awareness

Author (Year)	Methodology	Key Contributions	Hardware Gap
Redmon (2018)	YOLOv3 : Darknet-53 backbone for real-time detection.	Defined the SOTA for fast, task-agnostic object detection.	No inherent semantic understanding; detects all classes equally.
Gu et al. (2022)	ViLD : Uses CLIP to distill open-vocabulary knowledge.	Proved that small models can inherit "Teacher" intelligence.	Heavy Transformer overhead; not optimized for INT8/FPGA.
Han et al. (2021)	Dynamic Neural Networks : Survey of skip-connection/gating.	Established runtime sparsity to save power.	General theory; not applied to cross-modal task-alignment.
Zeng et al. (2022)	Socratic Models : LLM-based reasoning for vision.	Used text prompts to guide visual search results.	Post-processing only; doesn't optimize internal feature extraction.

03. Methodology: Dual-Stream Distillation

Our framework of **Knowledge Distillation** uses a "Teacher-Student" setup to move intelligence into a lightweight architecture.

1. Semantic Stream (Teacher)

- **CLIP Text Encoder:** Encodes "Slice" and "Cut" into identical vectors.
- **Tiny Task Mapper:** A distilled linear layer ($512 \times V$) that converts text into embeddings "on-the-go" for the FPGA.

2. Visual Stream (Student)

- **YOLOv8 Backbone:** Features tapped at Stage 4 ($20 \times 20 \times 512$).
- **Dynamic Gating:** The task vector modulates the image channels via **Sigmoid-Gated Hadamard Product**.

04. Novelty 1: Dynamic Gating Mechanism

Replacing **Heavy Attention** with **Lightweight Hardware-Native Gating**.

The Logic:

- **Full Attention:** $O(N^2)$ — Every pixel looks at the text. (Too heavy for VEGA).
- **Proposed Gating:** $O(C)$ — The task text creates 512 "Volume Knobs" applied globally to image channels.

Hardware Advantage:

By quantizing gates to **INT8**, the VEGA processor performs **Bit-Shift** operations to "mute" irrelevant channels, skipping 30-50% of multiplications in subsequent layers.

05. Novelty 2: Semantic Affordance Distillation

Training the Student (YOLO) to "think" like the Expert (CLIP).

The Process:

1. **Expert Opinion:** CLIP generates a 16×16 **Heatmap** indicating functional relevance (e.g., "Pouring").
2. **Student Alignment:** We interfere at the YOLO backbone, using a 1×1 **Convolution** and **Bilinear Interpolation** to match CLIP's dimensions.
3. **MSE Loss:** The Student adjusts its weights so its **Gated Feature Map** matches the Teacher's Semantic Heatmap.

Result: YOLO learns **Affordances** (use-cases) instead of just static labels.

06. Hardware Integration: Genesys-2 FPGA

Bridging the gap between High-Level Python and Low-Level RTL.

1. Quantization (INT8)

- **QAT:** Quantization-Aware Training ensures distillation remains stable with integers.
- **Precision:** Scaling from Float32 to 8-bit to fit VEGA ISA.

2. FPGA Execution

- **Task Vector:** Pre-encoded synonyms stored in on-chip BRAM.
- **Accelerator:** Custom logic for fast Hadamard Product and sparsity-skipping.

07. Expected Results & Metrics

We evaluate the system on three fronts: Accuracy, Intelligence, and Efficiency.

Metric	Target Score	Analysis
mAP (Detection)	> 0.45	Maintaining standard YOLOv8 detection accuracy.
Task Success Rate	> 85%	Percentage of times the task-relevant object is selected.
Inference Latency	< 20ms	Real-time performance on the VEGA Processor.
Power Savings	~30%	Reduction in power via Dynamic Gating/Sparsity.

08. Interdisciplinary Relevance (CS + ECE)

Computer Science Contribution:

- Advanced **Knowledge Distillation** and Cross-Modal Alignment.
- Natural Language Processing (NLP) for **Task Embedding and Synonyms**.

Electronics & Communication Contribution:

- **RTL Integration** of the VEGA Processor IP.
- Design of a hardware-efficient **Gating Accelerator** on FPGA.
- Low-level optimization for **RISC-V Instruction Set Architecture**.

09. Conclusion & Impact

1. **Innovation:** We pivot from "detecting all" to "**detecting what matters**" using a task-aware internal filter.
2. **Intelligence:** Knowledge Distillation allows a **smaller sized model** to exhibit the semantic reasoning of a **2GB VLM**.
3. **Hardware Native:** The **Gating Mechanism** is specifically designed for the Bit-Shift capabilities of the VEGA processor.
4. **Application:** Ideal for **Embedded Robotics** and **Smart Surveillance** where power and task-relevance are paramount.
5. **SDG 4 Alignment:** Provides a scalable solution for Quality Education, ensuring lab and classroom tools are accessible to every learner, regardless of visual ability.

Thank You
